**CHAPTER EXERCISE:** Answers for this chapter start on page 311.

**A calculator should NOT be used on the following questions.**

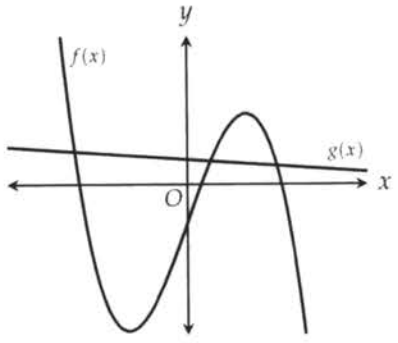
**1**

| $x$ | $y$ |
|---|---|
| 0 | 20 |
| 1 | 21 |
| 3 | 29 |

The table above displays several points on the graph of the function $f$ in the $xy$-plane. Which of the following could be $f(x)$ ?

A) $f(x) = 20x$
B) $f(x) = x + 20$
C) $f(x) = x - 20$
D) $f(x) = x^2 + 20$

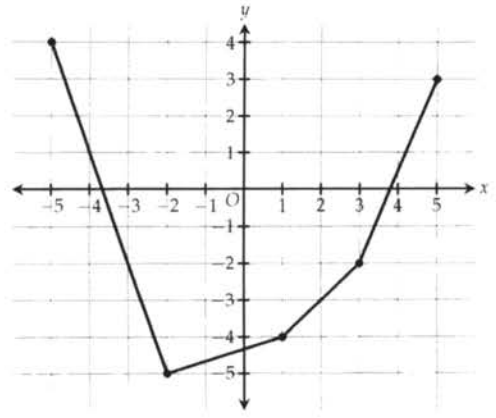
**2**




In the portion of the $xy$-plane shown above, for how many values of $x$ does $f(x) = g(x)$ ?

A) None
B) One
C) Two
D) Three

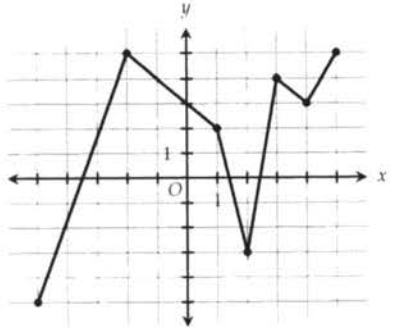
**3**




The graph of the function $f$ is shown in the $xy$-plane above. If $f(a) = f(3)$, which of the following could be the value of $a$ ?

A) $-4$
B) $-3$
C) $-2$
D) 1

**4**




The function $f$ is graphed in the $xy$-plane above. For how many values of $x$ does $f(x) = 3$ ?

A) Two
B) Three
C) Four
D) Five

**5**

For which of the following functions is it true that $f(-3) = f(3)$ ?

A) $f(x) = \frac{2}{x}$

B) $f(x) = \frac{x^3}{3}$

C) $f(x) = 3x^2 + 1$

D) $f(x) = x + 2$

**6**

The function $f$ is defined by $f(x) = 3x + 2$ and the function $g$ is defined by $g(x) = f(2x) - 1$. What is the value of $g(10)$ ?

**7**

If $f(x) = \frac{16 + x^2}{2x}$ for all $x \neq 0$, what is the value of $f(-4)$ ?

A) $-8$

B) $-4$

C) 4

D) 8

**8**

| $x$ | 0 | 1 | 2 |
|---|---|---|---|
| $f(x)$ | $-2$ | 3 | 18 |

Several values of the function $f$ are given in the table above. If $f(x) = ax^2 + b$ where $a$ and $b$ are constants, what is the value of $f(3)$ ?

A) 23

B) 39

C) 43

D) 56

**9**

If $f(x) = x^2$, for which of the following values of $c$ is $f(c) < c$ ?

A) $\frac{1}{2}$

B) 1

C) $\frac{3}{2}$

D) 2

**10**

If the graph of the function $f$ has $x$-intercepts at $-3$ and 2, and a $y$-intercept at 12, which of the following could define $f$ ?

A) $f(x) = (x+3)^2(x-2)$

B) $f(x) = (x+3)(x-2)^2$

C) $f(x) = (x-3)^2(x+2)$

D) $f(x) = (x-3)(x+2)^2$

**11**

$$f(x) = x^2 + 1$$

$$g(x) = x^2 - 1$$

The functions $f$ and $g$ are defined above. What is the value of $f(g(2))$ ?

A) 3

B) 5

C) 10

D) 17

**12**

In the $xy$-plane, which of the following translations of the graph of $y = 2x^2 - 2$ results in the graph of $y = 2x^2 + 4$ ?

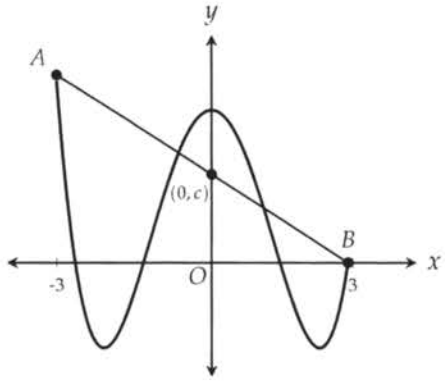
A) A translation 2 units downward

B) A translation 6 units upward

C) A translation 2 units to the left

D) A translation 6 units to the right

**13**




The graph of the function $f$ and line segment $\overline{AB}$ are shown in the $xy$-plane above. For how many values of $x$ between $-3$ and 3 does $f(x) = c$ ?

**14**

| $x$ | $f(x)$ |
|---|---|
| $-4$ | 3 |
| $-2$ | 5 |
| 0 | 2 |
| 2 | 16 |
| 3 | 4 |
| 4 | 8 |

The table above gives some values for the function $f$. If $g(x) = 2f(x)$, what is the value of $k$ if $g(k) = 8$ ?

A) 2
B) 3
C) 4
D) 8

**15**

In the $xy$-plane, the graph of the function $f$ reaches its maximum value at the point $(3, f(3))$. The function $g$ is defined by $g(x) = f(x) + 7$. At which of the following points in the $xy$-plane does the graph of $g$ reach its maximum value?

A) $(10, f(10) + 7)$
B) $(f(3), f(3) + 7)$
C) $(3, f(10))$
D) $(3, f(3) + 7)$

**16**

$$f(x) = \sqrt{x - 2}$$

The function $f$ is defined above for all $x \geq 2$. Which of the following is equal to $f(18) - f(11)$ ?

A) $f(3)$
B) $f(5)$
C) $f(6)$
D) $f(7)$

**17**

The function $f$ is defined by $f(x) = (x - 3)^2$, and the function $g$ is defined by $g(x) = x^2 + 4x + 4$. The graph of $g$ in the $xy$-plane is the graph of $f$ shifted $k$ units to the left. What is the value of $k$ ?

**A calculator is allowed on the following questions.**

**18**

$$y = \frac{x+1}{x-1}$$

Which of the following points in the $xy$-plane is NOT on the graph of $y$ ?

A) $(-2, \frac{1}{3})$

B) $(-1, 0)$

C) $(0, -1)$

D) $(1, 2)$

**19**

Let the function $g$ be defined by $g(x) = \sqrt{3x}$. If $g(a) = 6$, what is the value of $a$ ?

A) 3

B) 6

C) 9

D) 12

**Questions 20-21 refer to the following information.**

| $x$ | $f(x)$ | $g(x)$ |
|---|---|---|
| $-2$ | 3 | 4 |
| $-1$ | 5 | 2 |
| 0 | $-2$ | $-3$ |
| 1 | 3 | 5 |
| 2 | 6 | 7 |
| 3 | 7 | 1 |

The functions $f$ and $g$ are defined for the six values of $x$ shown in the table above.

**20**

What is the value of $f(g(-1))$ ?

A) $-2$

B) 3

C) 5

D) 6

**21**

If $g(c) = 5$, what is the value of $f(c)$ ?

A) $-2$

B) 3

C) 5

D) 6

**22**

If $f(x) = -3x + 5$ and $\frac{1}{2}f(a) = 10$, what is the value of $a$ ?

A) $-8$

B) $-5$

C) 5

D) 8

**23**

| $x$ | $f(x)$ |
|---|---|
| 0 | $-4$ |
| 1 | $-8$ |
| 2 | 3 |
| 3 | 6 |
| 4 | 7 |
| 5 | 2 |
| 6 | 4 |
| 7 | 5 |

Several values of the function $f$ are given in the table above. If the function $g$ is defined by $g(x) = f(2x - 1)$, what is the value of $g(3)$ ?

A) 2

B) 6

C) 5

D) 7

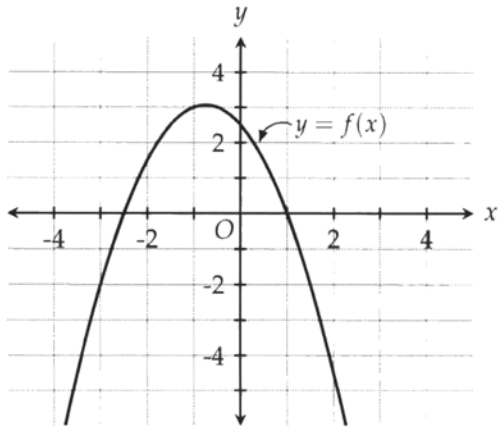
**24**

$$f(x) = 4x - 3$$
$$g(x) = 3x + 5$$

The functions $f$ and $g$ are defined above. Which of the following is equal to $f(8)$ ?

A) $g(1)$
B) $g(3)$
C) $g(5)$
D) $g(8)$

**25**




The graph of $f(x)$ is shown in the $xy$-plane above. If $g(x) = (x + 3)(x - 1)$, for which of the following values of $x$ is $f(x) > g(x)$ ?

A) $-3$
B) $-2$
C) 1
D) 2

**26**

In the $xy$-plane, the graph of the function $g$ is the image of the graph of the function $f$ after a translation of $1\frac{1}{2}$ units to the right. Which of the following defines $g(x)$ ?

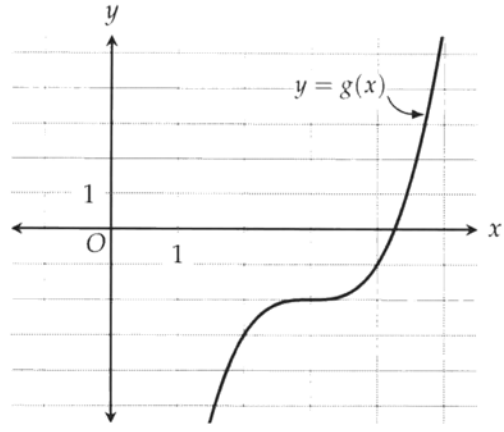
A) $g(x) = f(3x - 2)$
B) $g(x) = f(3x + 2)$
C) $g(x) = f(2x - 3)$
D) $g(x) = f(2x + 3)$

**27**

If $f(x)$ is a linear function such that $f(2) \leq f(3)$, $f(4) \geq f(5)$, and $f(6) = 10$, which of the following must be true?

A) $f(3) < f(0) < f(4)$
B) $f(0) = 0$
C) $f(0) > 10$
D) $f(0) = 10$

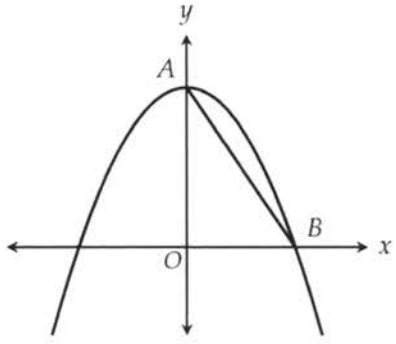
**28**




The graph of the function $g$ is shown in the $xy$-plane above, and the function $f$ (not shown) is defined by $f(x) = x^3$. If $g$ is defined by $g(x) = f(x + a) + b$, where $a$ and $b$ are constants, what is the value of $a + b$ ?

A) $-5$
B) $-1$
C) 1
D) 5

**29**




The graph of the function $y = 9 - x^2$ is shown in the $xy$-plane above. What is the length of $\overline{AB}$ ?

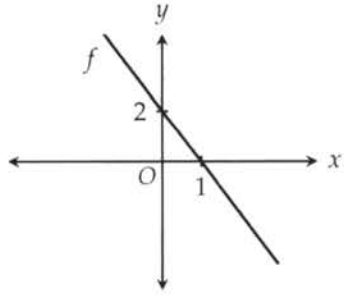
A) $3\sqrt{2}$

B) $3\sqrt{10}$

C) 9

D) $9\sqrt{10}$

**30**




The function $f$ is graphed in the $xy$-plane above. If the function $g$ is defined by $g(x) = f(x) + 4$, what is the $x$-intercept of $g(x)$ ?

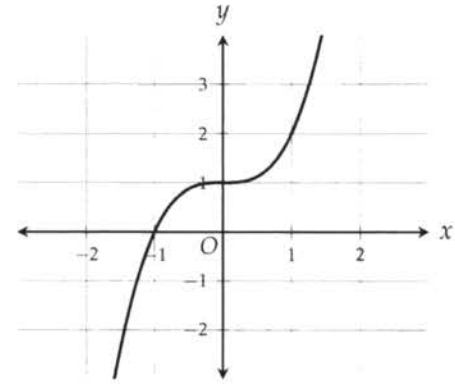
A) $-3$

B) $-1$

C) 3

D) 4

**31**




The function $f(x) = x^3 + 1$ is graphed in the $xy$-plane above. If the function $g$ is defined by $g(x) = x + k$, where $k$ is a constant, and $f(x) = g(x)$ has 3 solutions, which of the following could be the value of $k$ ?

A) $-1$

B) 0

C) 1

D) 2

**32**

In the $xy$-plane, the function $y = ax + 12$, where $a$ is a constant, passes through the point $(-a, a)$. If $a > 0$, what is the value of $a$ ?

# Chapter 16: Functions

**CHAPTER EXERCISE:**

1. [D] Check each answer choice to see whether $f(0) = 20, f(1) = 21$, and $f(3) = 29$. The only function that satisfies all three is (D).
2. [D] $f(x) = g(x)$ when the two graphs intersect. They intersect at 3 points, so there must be 3 values of $x$ where $f(x) = g(x)$.
3. [B] $f(3) = -2$. Now where else is $f$ at $-2$? When $x = -3$. So $a$ must be $-3$.
4. [C] Draw a horizontal line at $y = 3$. This line intersects $f(x)$ four times, so there are four solutions (four values of $x$ for which $f(x) = 3$).
5. [C] Plug in $-3$ and 3 into each of the answer choices to see whether you get the same value. If you're smart about it, you'll realize that answer (C) has an $x^2$, which always gives a positive value. Testing (C) out, $f(-3) = 3(-3)^2 + 1 = 28$ and $f(3) = 3(3)^2 + 1 = 28$. The answer is indeed (C).
6. [61] First, $g(10) = f(20) - 1$. Now, $f(20) = 3(20) + 2 = 62$. Finally, $g(10) = 62 - 1 = 61$.
7. [B] $f(-4) = \dfrac{16 + (-4)^2}{2(-4)} = \dfrac{32}{-8} = -4$.
8. [C] We plug in values to solve for $a$ and $b$. Plugging in $(0, -2)$, $-2 = a(0)^2 + b = b$. So, $b = -2$. Plugging in $(1, 3)$,

$$3 = a(1)^2 + b$$
$$3 = a - 2$$
$$5 = a$$

So $a = 5$ and $f(x) = 5x^2 - 2$. Finally, $f(3) = 5(3)^2 - 2 = 43$.

9. [A] Plug in the answer choices and check. $f\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$, which is less than $\frac{1}{2}$. The answer is (A).
10. [B] The $x$-intercepts of $-3$ and 2 mean that $f(x)$ must have factors of $(x + 3)$ and $(x - 2)$. That eliminates (C) and (D). A $y$-intercept of 12 means that when we plug in $x = 0$, $f(x) = 12$. Only answer (B) meets all these conditions.
11. [C] $g(2) = 2^2 - 1 = 3$. So, $f(g(2)) = f(3) = 3^2 + 1 = 10$.
12. [B] The difference between $2x^2 - 2$ and $2x^2 + 4$ is a constant of 6. In other words, 6 needs to be added on to $y = 2x^2 - 2$ to get $y = 2x^2 + 4$. That entails a translation 6 units upward.
13. [3] Draw a horizontal line at $y = c$, passing through $(0, c)$. This horizontal line intersects with $f$ three times. That means there are 3 values of $x$ for which $f(x) = c$.
14. [B]

$$g(k) = 8$$
$$2f(k) = 8$$
$$f(k) = 4$$

Looking at the chart, $f(x) = 4$ only when $x = 3$. So $k = 3$.

15. [D] Since $g(x)$ just adds a constant of 7 to every value of $f(x)$, the maximum of $g(x)$ must occur at the same $x$-value as the maximum of $f(x)$, namely $x = 3$. So, the maximum of $g(x)$ is reached at the point $(3, g(3))$, and since $g(x) = f(x) + 7$, this point is $(3, f(3) + 7)$.
16. [A] $f(18) = \sqrt{18 - 2} = \sqrt{16} = 4$. $f(11) = \sqrt{11 - 2} = \sqrt{9} = 3$. $f(18) - f(11) = 4 - 3 = 1$. Testing each answer choice, $f(3)$ is the only one that also equals 1.

17. [5] If we factor $g$, we get $g(x) = x^2 + 4x + 4 = (x+2)^2$. Since $g(x)$ is $f(x)$ shifted $k$ units to the left,

$$\begin{aligned} g(x) &= f(x+k) \\ (x+2)^2 &= (x+k-3)^2 \\ x+2 &= x+k-3 \\ 2 &= k-3 \\ 5 &= k \end{aligned}$$

Alternatively, we could've solved this question by comparing $f(x)$ and $g(x)$ to $y = x^2$. The graph of $f(x) = (x-3)^2$ is 3 units to the right of $y = x^2$, and the graph of $g(x) = (x+2)^2$ is 2 units to the left of $y = x^2$. Therefore, $g(x)$ is 5 units to the left of $f(x)$.

18. [D] $(1,2)$ cannot be on the graph of $y$ since an $x$-value of 1 would result in division by 0.

19. [D]

$$\begin{aligned} g(a) &= 6 \\ \sqrt{3a} &= 6 \\ 3a &= 36 \\ a &= 12 \end{aligned}$$

20. [D] Using the table, $g(-1) = 2$. Then, $f(2) = 6$.

21. [B] If $g(c) = 5$, then $c = 1$ since 1 is the only input that gives an output of 5. Then, $f(c) = f(1) = 3$.

22. [B] From the second equation, $f(a) = 20$. So,

$$\begin{aligned} f(a) &= -3a + 5 \\ 20 &= -3a + 5 \\ 3a &= -15 \\ a &= -5 \end{aligned}$$

23. [A] $g(3) = f(2(3)-1) = f(5) = 2$. We get $f(5) = 2$ from the table.

24. [D] $f(8) = 4(8) - 3 = 29$. Testing each answer choice to see which one yields 29, we see that $g(8) = 3(8) + 5 = 29$.

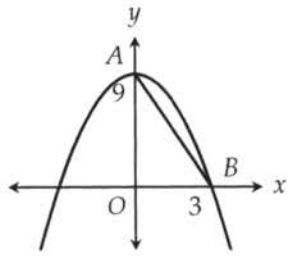
25. [B] Test each of the answer choices. When $x = -3$, $f(x) = -2$ according to its graph, and $g(x) = (-3+3)(-3-1) = 0$. In this case, $f(x)$ is not greater than $g(x)$. When $x = -2$, $f(x) \approx 1.5$ and $g(x) = (-2+3)(-2-1) = -3$. In this case, $f(x) > g(x)$ so we have our answer.

26. [C] For horizontal shifts, the trick is to find the value of $x$ that makes the substituted expression equal to 0. This value tells you what the horizontal shift is. For choice A, $x = \frac{2}{3}$ makes $3x - 2$ equal to 0, so the horizontal shift is $\frac{2}{3}$ units to the right. For choice B, that value is $-\frac{2}{3}$, so the shift is $\frac{2}{3}$ units to the left. For choice C, $x = \frac{3}{2}$ makes the expression $2x - 3$ equal to 0, so the shift is $\frac{3}{2}$ units to the right. This is the answer. For choice D, the shift is $\frac{3}{2}$ units to the left.

27. [D] The key words are "linear function": $f$ is a straight line. So for what straight line can both $f(2) \leq f(3)$ and $f(4) \geq f(5)$ be true? Only a horizontal straight line. Take a minute to think that through. Now, since $f$ is a flat line and $f(6) = 10$, then all values of $f$ are 10, no matter what the value of $x$ is. Therefore, $f(0) = 10$.

28. [A] Remember that you can use your calculator for graphing. The graph of $f(x) = x^3$ is "centered" at $(0,0)$. The graph of $g(x)$ is "centered" at $(3,-2)$. Comparing these points of reference, we can see that $g(x)$ is shifted 3 units to the right and 2 units downward from $f(x)$. Therefore, $g(x) = f(x-3) - 2$, which means $a = -3$ and $b = -2$. The sum $a + b$ is then $-3 + (-2) = -5$.

29. [B] When $x = 0$, $y = 9$, so the $y$-intercept is 9. When $y = 0$, $x = 3$, so the $x$-intercept is 3.

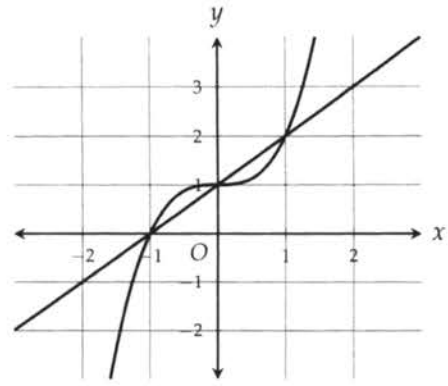



$\triangle AOB$ is a right triangle with a base of 3 and a height of 9. Using the pythagorean theorem,

$$AO^2 + OB^2 = AB^2$$
$$9^2 + 3^2 = AB^2$$
$$90 = AB^2$$
$$3\sqrt{10} = AB$$

30. [C] The graph of $g$ is 4 units up from where $f$ is, but because the slope of $f$ is $-2$, the $x$ and $y$ intercepts of $g$ will not increase by the same amount. They'll increase in a ratio of 2:1. So when the $y$-intercept gets shifted up by 4, the $x$-intercept gets shifted to the right by 2. The new $x$-intercept is therefore $1 + 2 = 3$. Another way to do this is to actually solve for the $x$-intercept. Using slope-intercept form, we get $f(x) = -2x + 2$. Adding 4 to get the equation of $g$, $g(x) = -2x + 6$. Setting $g(x) = 0$ and solving for $x$ to get the $x$-intercept, we get $x = 3$.

31. [C] The function $g(x)$ is a line with a slope of 1 and a $y$-intercept of $k$. If you draw $g(x)$ with the different possibilities for $k$ from the answer choices, you'll see that there's an intersection of 3 points with $f(x)$ only when $k = 1$ as shown below.




32. [3] Plugging in $(-a, a)$,

$$a = a(-a) + 12$$
$$a = -a^2 + 12$$
$$a^2 + a - 12 = 0$$
$$(a + 4)(a - 3) = 0$$
$$a = -4, 3$$

Since $a > 0$, $a = 3$.